

# Design and Implementation of a Web System for Searching and Managing Academic Courses

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**Abstract**—Efficient access to academic information has become a challenge for institutions and users seeking courses that meet their needs. In this context, we present the development of a web application called Course Search, which aims to facilitate course consultation and administration through an intuitive interface and modular design.

The proposed solution integrates a search service capable of filtering and classifying courses according to defined criteria, along with an administration module exclusively for the administrator user, who manages the registered courses and teachers. The system was designed following service-oriented architecture principles, using UML and activity diagrams that support the clarity of the design and the correct structuring of the system.

**Index Terms**—course search engine, web systems, academic management, service-oriented modular architecture, process optimization.

## I. INTRODUCTION

In the context of today's higher education, internal management of academic information represents a constant challenge for institutions. In many cases, students and professors at the same university face difficulties when trying to locate available courses or identify who teaches them, because the data is scattered across different platforms or institutional documents. This lack of centralization limits efficiency in consultation and hinders academic planning.

In this scenario, there is a need to develop a specialized search system for educational institutions that allows quick and organized access to academic offerings. This system aims to optimize interaction between users and the university's internal information, improving transparency, accessibility, and the search experience.

Previous research on academic information management and search systems has focused on improving access to course-related data within educational institutions. For example, the University of Tartu developed an online study information system that allows students and staff to browse, filter, and retrieve detailed course information through a centralized institutional platform [1]. This system exemplifies how universities can use digital infrastructures to centralize course data and

make it easily accessible to their academic communities. Similarly, commercial solutions such as the Terminalfour Course Search module provide customizable search engines for higher education institutions, enabling students to explore available programs and subjects based on filters such as academic area or level of study [2].

While both initiatives enhance discoverability and transparency of academic offerings, they remain oriented toward course visibility and student decision-making rather than modeling the structural relationship between courses and the professors who teach them. Furthermore, most existing solutions are designed to serve multiple departments or institutions through generalized catalog structures. In contrast, the present work focuses on a single academic institution and emphasizes the bidirectional association between courses and instructors, providing a more specific and administratively useful search system for institutional management and academic organization.

The work presented below is organized as follows: Section II describes the methodology and architecture used in developing the system; Section III presents the results obtained and their analysis; Section IV presents the conclusions and possible lines of future work.

## II. METHODS AND MATERIALS

The development of the proposed system is based on a layered architecture, designed to ensure a clear separation of responsibilities, promote maintainability, and facilitate the future scalability of the project. This structure allows the system components to be organized logically, so that each layer fulfills a specific purpose within the information flow, from user interaction to persistent data management.

The presentation layer corresponds to the environment where users interact, offering an intuitive and functional interface that allows them to search for courses and teachers within an educational institution. This layer communicates with the business logic through controlled requests, preventing direct access to data and preserving the integrity of operations.

The interface has been designed with the aim of offering a fluid experience, prioritizing accessibility and clarity in navigation.

The business logic layer concentrates the main rules of the system and defines the processes that enable coordination of operations between different entities. This layer includes components such as AuthService, which manages administrator authentication; SearchService, which processes course and professor searches; and ProfessorCourseManager, which maintains the relationship between teachers and subjects. These classes act as mediators between the presentation and the data, ensuring that each user request is validated and processed correctly before accessing the stored information.

The data layer is responsible for the persistence and retrieval of information. It consists of the Course and Professor entities, which represent the fundamental objects of the academic domain. The relationship between the two entities is managed through associations controlled by the business layer, ensuring the consistency of registration, search, and update operations. This design facilitates the future extension of the model, allowing new attributes or entities to be included without compromising the overall stability of the system.

In terms of design, the class diagram illustrates the structural organization of the system and the interaction between the main components. Layered architecture promotes loose coupling and high cohesion, two essential principles for achieving modular and maintainable software. In this way, each layer can evolve independently, allowing, for example, the user interface to be replaced or improved without altering the business rules or data structure.

In addition, the model was designed with clarity and reproducibility in mind. All design decisions were made seeking a balance between simplicity and functionality, so that the implementation can be replicated or extended by other developers under the same conceptual premises.

Finally, the restriction to a single educational institution and the lack of integration with external systems are recognized as initial limitations, aspects that may be addressed in later versions of the project.

### III. RESULTS & DISCUSSION

### IV. CONCLUSIONS

### REFERENCES

- [1] U. of Tartu. (2015) Access to course information: The online study information system at the university of tartu. Erasmus+ European Commission. Accessed: 2025-10-20. [Online]. Available: <https://erasmus-plus.ec.europa.eu/eche/access-to-course-information-the-online-study-information-system-at-the-university-of-tartu>

- [2] Terminalfour. (2023) Course search solutions for higher education. Terminalfour. Accessed: 2025-10-20. [Online]. Available: <https://www.terminalfour.com/solutions/course-search/>